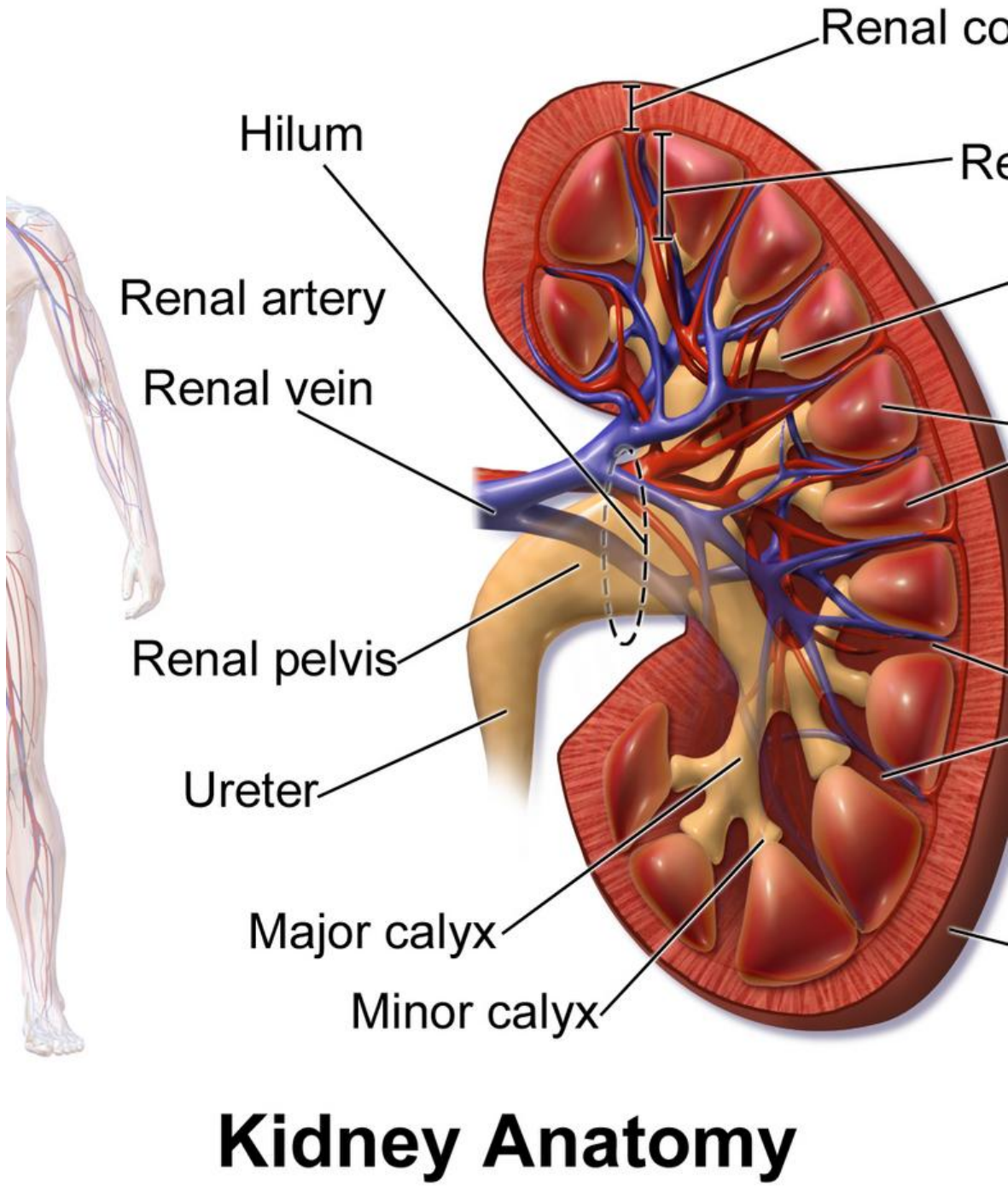
DP Biology – D3.3 (HL extension)

The kidney and blood flow

- How do humans keep internal conditions constant?
- Why is it useful for organisms to keep internal conditions constant?

HL statements D3.3.7–11. Textbook pp. 768–778.
Image: the kidney. Blausen Medical, CC BY 3.0.



Objectives for each class

Class	Statements	By the end of the class, you can...
1	D3.3.7–8	Distinguish excretion from osmoregulation, and explain ultrafiltration and selective reabsorption
2	D3.3.9–10	Explain how the loop of Henle and ADH let the kidney control water loss
3	D3.3.11	Explain how blood supply to organs changes with activity

Two jobs of the kidney

D3.3.7

Excretion

Removing waste made by the body's own metabolism, such as urea from breaking down amino acids.

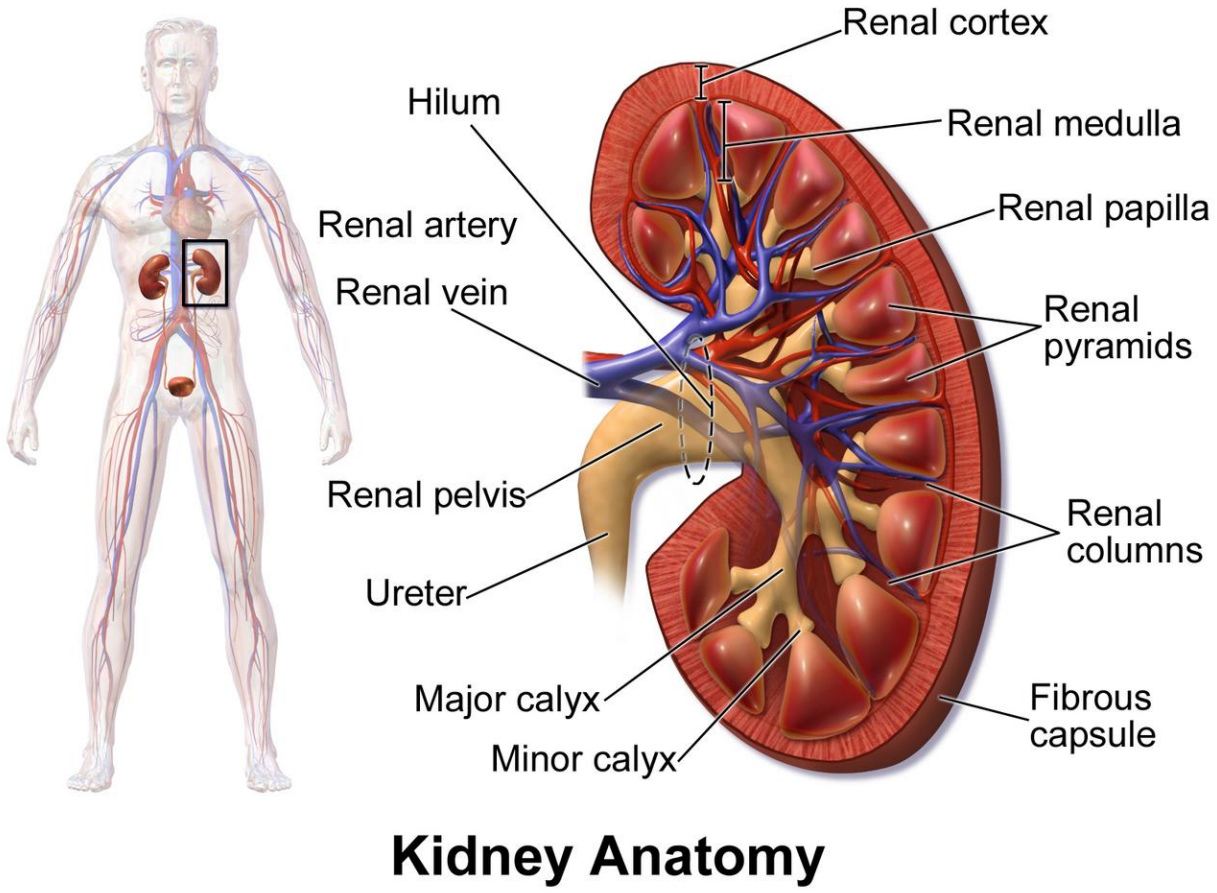
Osmoregulation

Keeping the water and solute concentration of the blood constant.

Why both matter

Urea is toxic if it builds up. If blood gets too concentrated, cells lose water by osmosis; if too dilute, they swell. The kidney fixes both at once by adjusting what goes into urine.

The kidney



Blood enters in the **renal artery** and leaves cleaned in the **renal vein**.

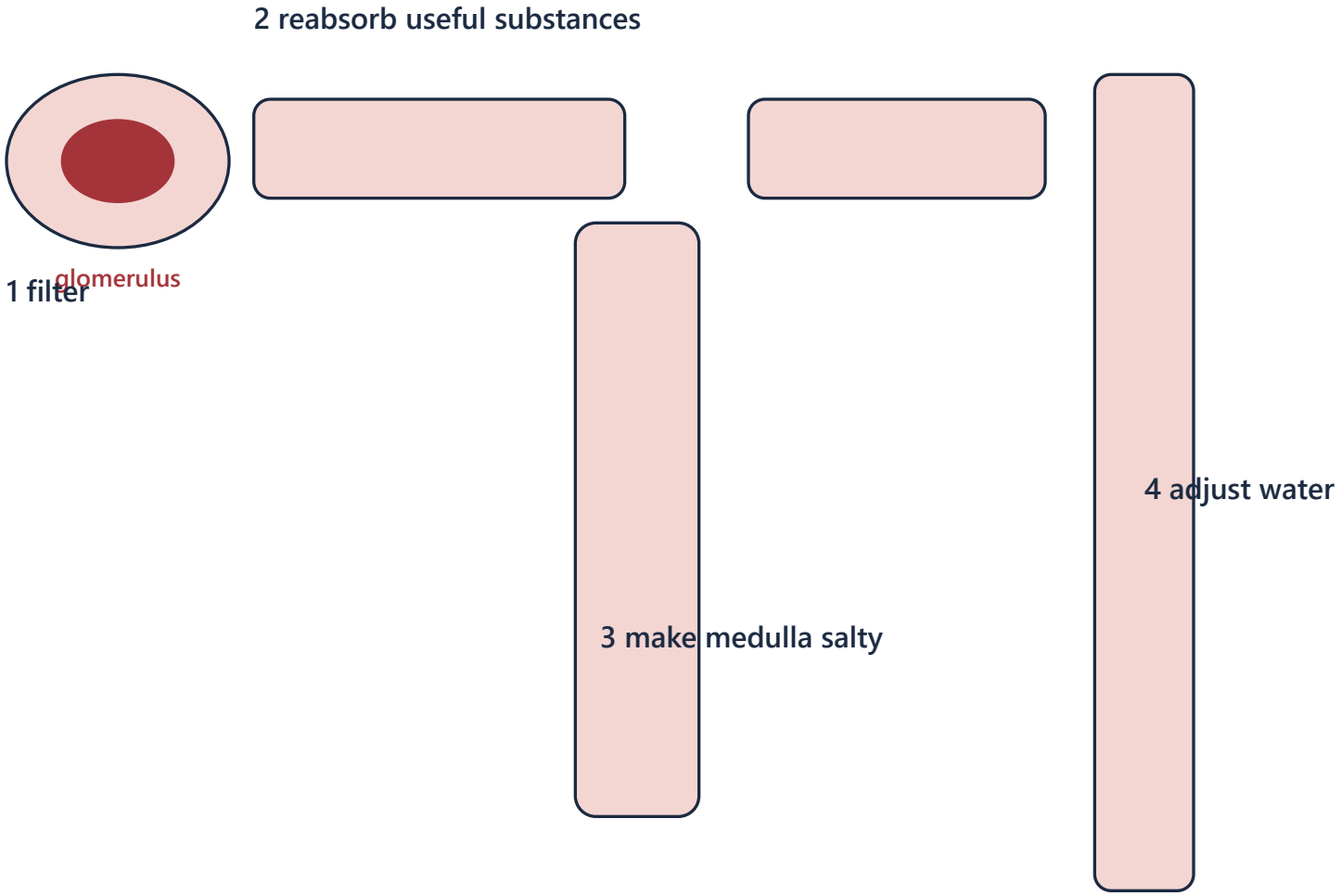
The outer **cortex** and inner **medulla** hold about a million **nephrons**: the tiny tubes that make urine.

Urine drains through the **ureter** to the bladder.

The kidney. Blausen Medical, CC BY 3.0.

The nephron

D3.3.8



Each nephron works in stages:

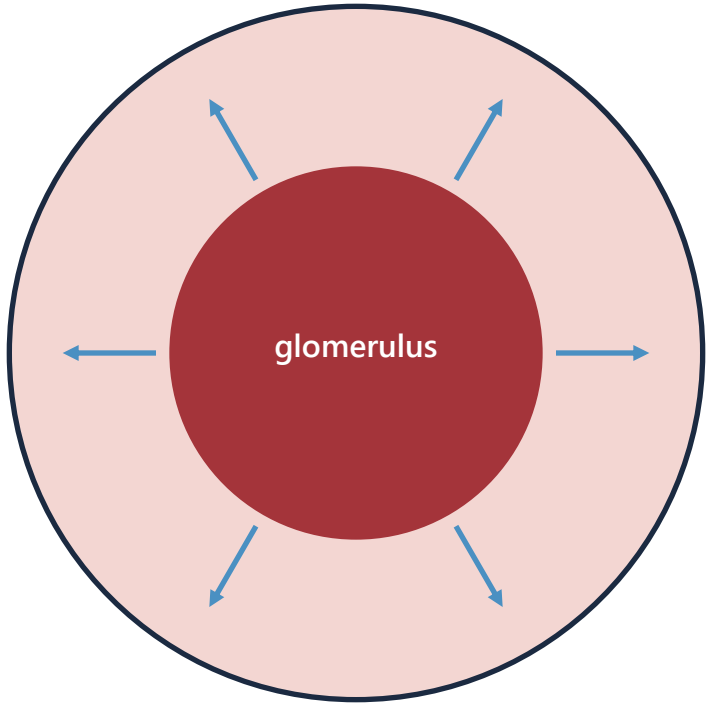
Filter the blood.

Reabsorb what the body needs.

Concentrate the urine using the loop of Henle.

Adjust how much water leaves, using ADH.

Ultrafiltration in the glomerulus



Bowman's capsule

Blood arrives in the **glomerulus** at **high pressure**: the vessel leaving it is narrower than the one entering.

Water and small molecules are forced through three layers (capillary wall, basement membrane, podocytes) into the capsule: **ultrafiltration**.

Cells and proteins are too big and stay in the blood.

What gets through?

D3.3.8

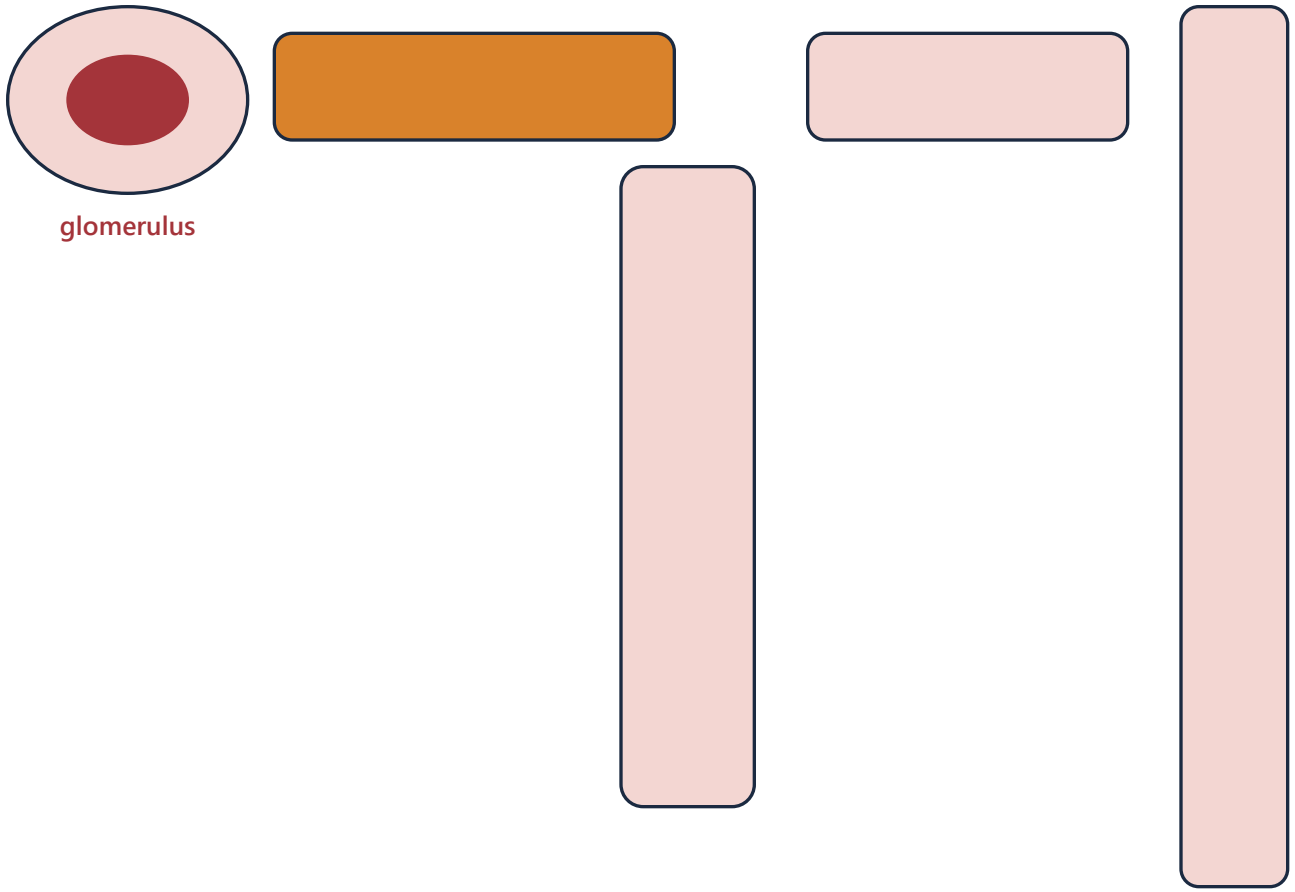
Substance	In blood plasma	In filtrate
Water, glucose, amino acids	yes	yes
Urea, salts	yes	yes
Plasma proteins	yes	almost none
Red blood cells	yes	none

The problem

The filter lets glucose and amino acids out along with the waste. About 180 litres of filtrate are made a day, but only about 1.5 litres of urine. Almost everything must be taken back.

Taking back what is useful

D3.3.8



In the **proximal convoluted tubule**, all the glucose and amino acids, and most water and salts, are taken back into the blood: **selective reabsorption**.

Its cells have **microvilli** and many **mitochondria** to power active transport.

Working with data: what the nephron keeps

D3.3.8

Amounts filtered and excreted per day (rounded typical values).

Substance	Filtered	In urine
Water / litres	180	1.5
Glucose / g	180	0
Urea / g	56	28
Sodium / g	600	4

1. Calculate the % of water reabsorbed.
2. Explain why no glucose is in the urine.
3. Explain why half the urea is excreted.

Check-in: class 1

D3.3.7–8

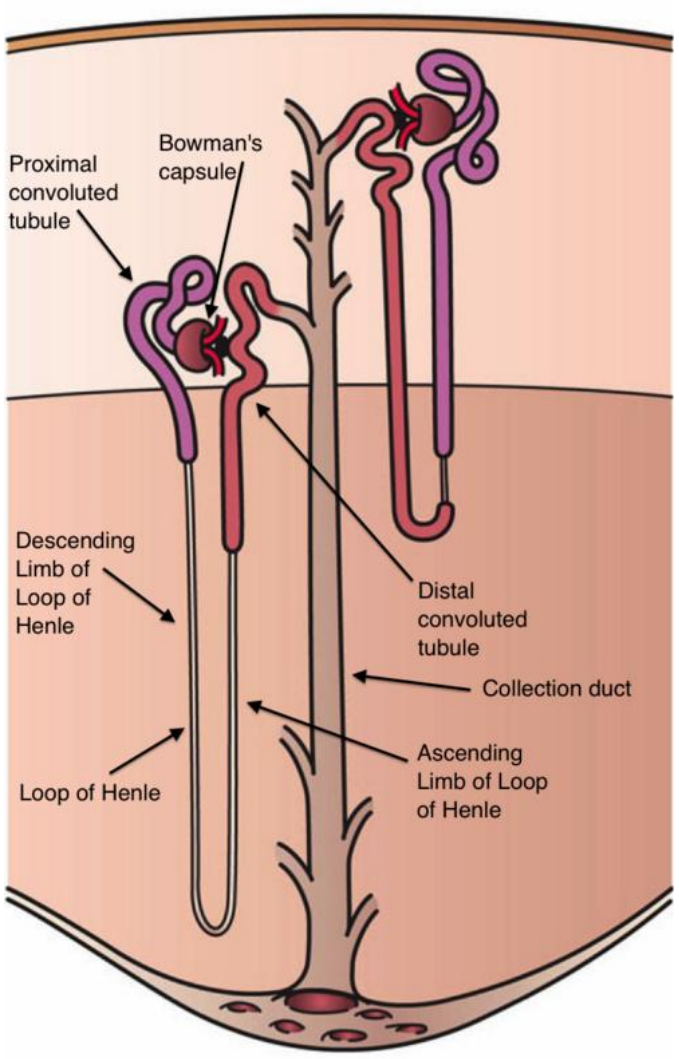
1. Distinguish between excretion and osmoregulation.
2. Explain why blood pressure in the glomerulus is high.
3. State two substances that are filtered and two that are not.
4. Define selective reabsorption.
5. Explain how proximal tubule cells are adapted.

How to use this

Answer without your notes, then check against slides 3–9.

Built for reabsorption

D3.3.8

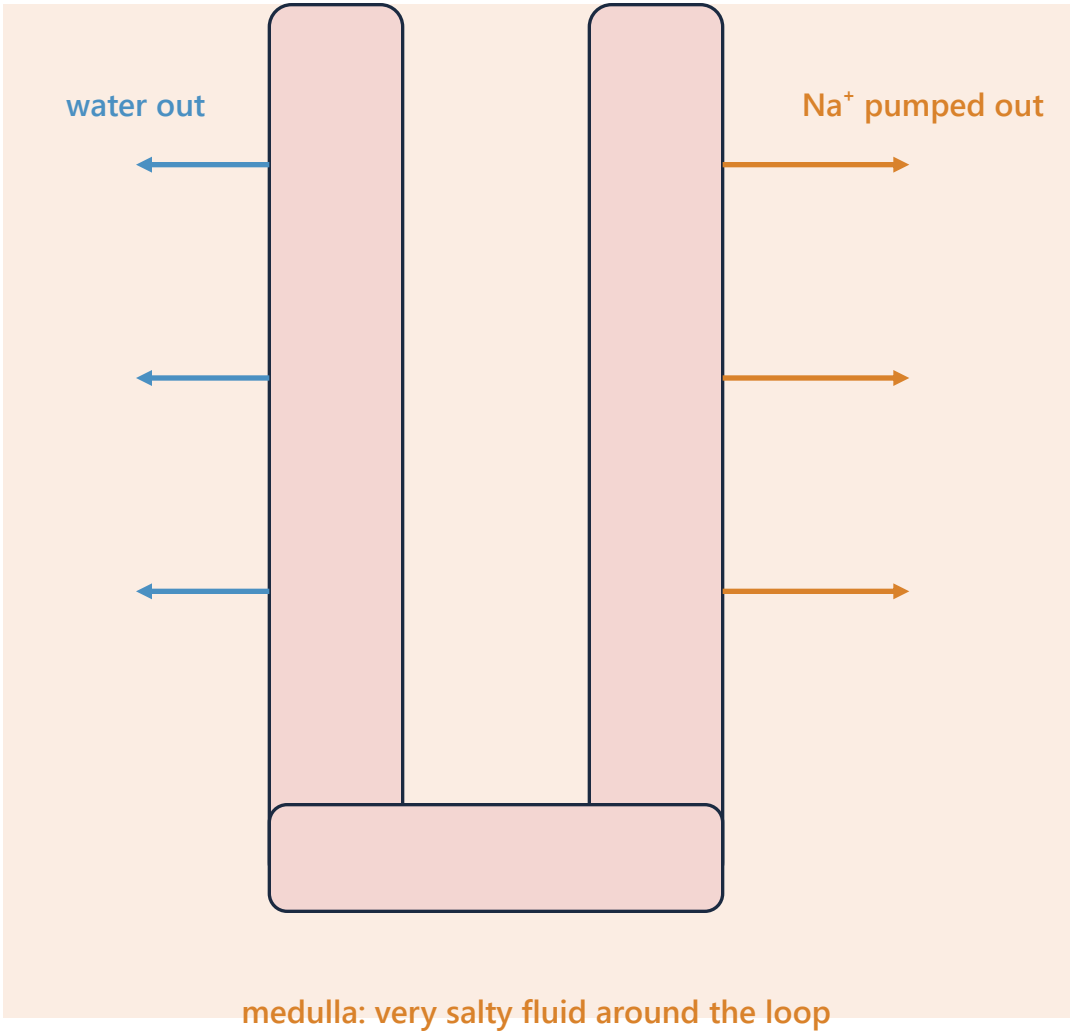


Kidney nephron. CC BY 3.0.

Trace the nephron in the diagram: capsule, proximal tubule, loop, distal tubule, collecting duct.

Each part has cells suited to its job. The proximal tubule does most of the work, reabsorbing about two thirds of the water.

The loop of Henle



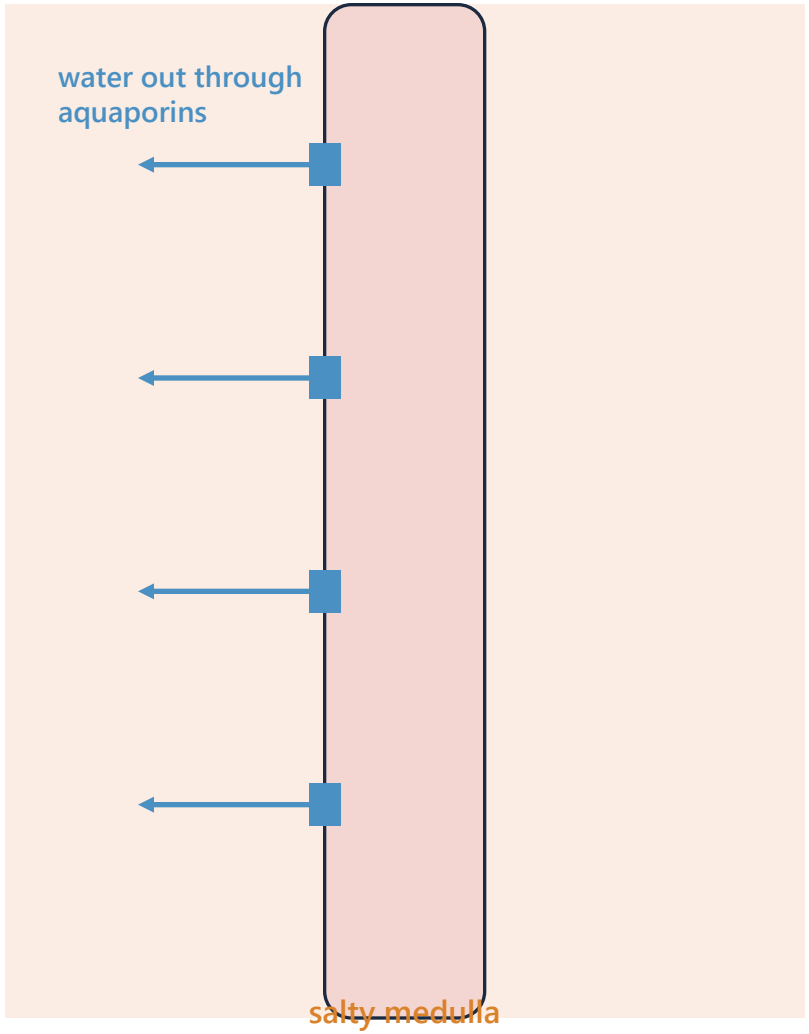
The **ascending limb** pumps Na^+ out into the medulla but does not let water follow.

This makes the medulla very **salty**.

The **descending limb** lets water out by osmosis into that salty fluid.

Result: a steep salt gradient in the medulla that the collecting duct uses later.

ADH and the collecting duct



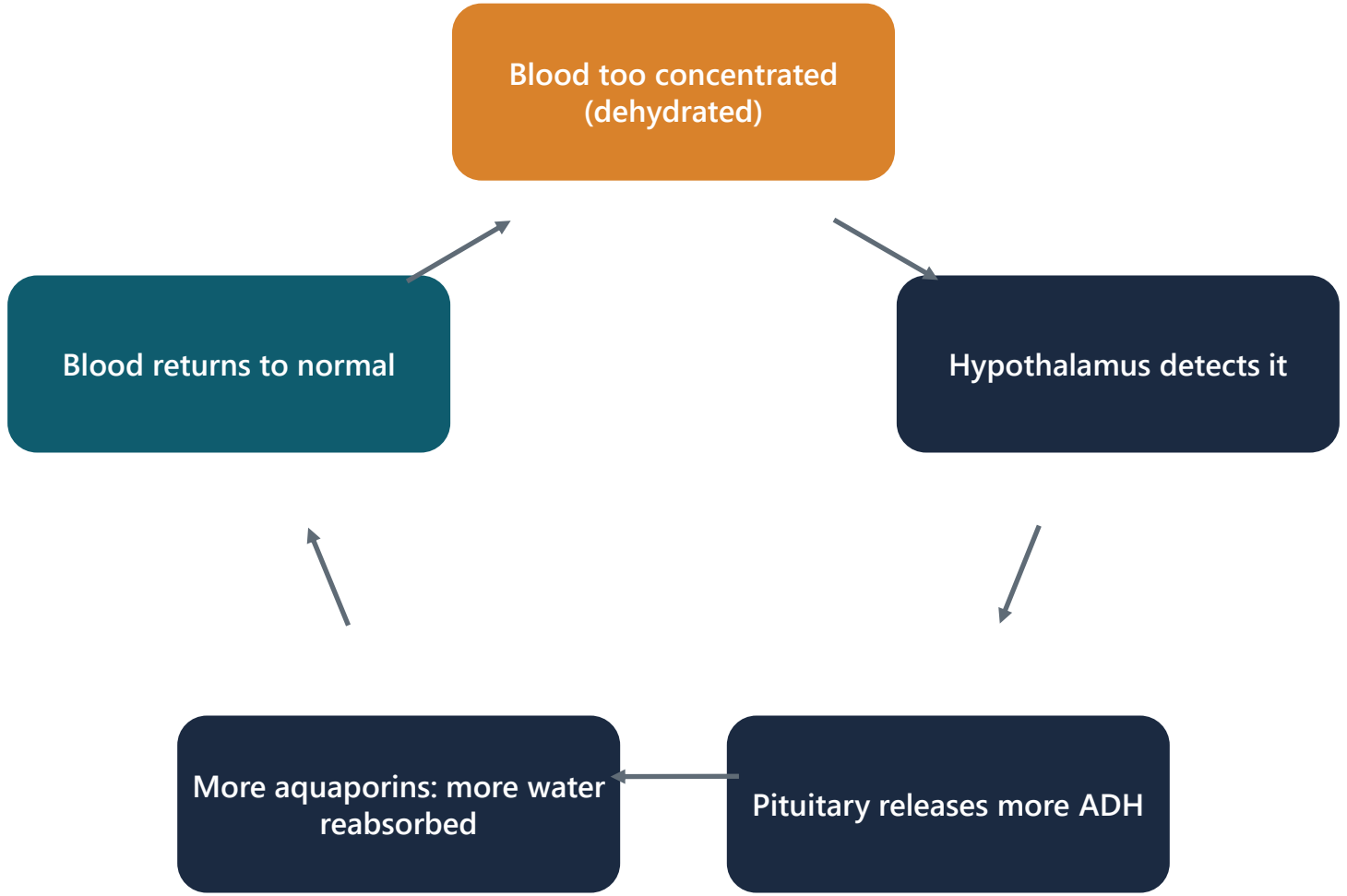
ADH (antidiuretic hormone) from the pituitary makes collecting duct cells put more **aquaporins** into their membranes.

More water leaves the duct by osmosis into the salty medulla: less, more concentrated urine.

No ADH: few aquaporins, lots of dilute urine.

Osmoregulation by negative feedback

D3.3.10



Working with data: drinking and urine

D3.3.10

Urine volume and concentration after a student drank 1 litre of water. These data are invented for practice; the pattern is realistic.

Time / min	Volume / cm^3 per 15 min	Concentration
0	50	high
30	220	low
60	300	very low
120	90	normal

1. Describe the changes.
2. Explain them in terms of ADH and aquaporins.
3. Predict the results if the student had sweated heavily instead.

Practice: explain the urine

D3.3.9–10

Kangaroo rat

Very long loops of Henle; never drinks.

After a big party

Alcohol lowers ADH release.

Hot day, no water

A runner in the heat.

1. For each, predict the volume and concentration of urine.
2. Explain using loop of Henle, ADH or aquaporins.

Check-in: class 2

D3.3.9–10

1. Explain how the ascending limb makes the medulla salty.
2. Explain why water leaves the descending limb.
3. State where ADH is released from.
4. Explain how ADH changes the collecting duct.
5. Describe the feedback loop when you are dehydrated.

How to use this

Answer without your notes, then check against slides 12–16.

The nephron in one table

D3.3.7–10

Part	What happens
Glomerulus and capsule	ultrafiltration: water and small molecules forced out under pressure
Proximal tubule	selective reabsorption: all glucose, amino acids; most water and salt
Loop of Henle	makes the medulla salty
Collecting duct	water reabsorbed by osmosis, controlled by ADH

Where does the blood go?

D3.3.11

Organ	At rest / %	Exercise / %
Skeletal muscle	20	80
Gut	25	4
Kidneys	20	3
Brain	15	4
Skin	5	6

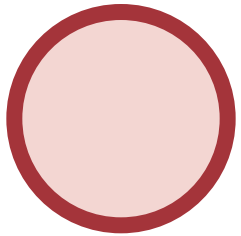


Athlete training. Florida Memory, public domain.

Values are percentages of the heart's output; total output rises about 5 times in hard exercise. Rounded typical values.

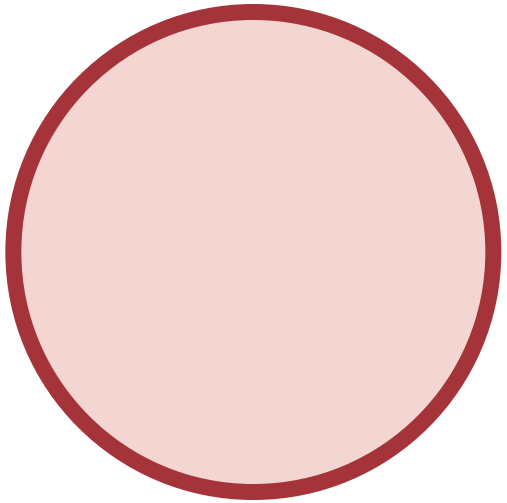
How blood is redirected

D3.3.11



less blood

arteriole wall contracted: vasoconstriction



arteriole wall relaxed: vasodilation

Arterioles have smooth muscle in their walls. To muscles in exercise they **dilate**, letting more blood through; to the gut and kidneys they **constrict**, letting less through.

Signals come from nerves, adrenaline and local chemicals (CO_2, low O_2) in active tissue.

Blood flow and temperature

D3.3.11



Infrared image: warm skin shows bright. Public domain.

When you are hot, arterioles to the **skin** dilate: more heat is lost and you go red.

When cold, they constrict: blood stays deep in the body and skin looks pale.

Link

Thermoregulation (D3.3 SL): the same arterioles adjust blood flow for temperature as well as activity.

Working with data: blood flow in exercise

D3.3.11

Blood flow to organs (cm^3 per minute). Rounded typical values.

Organ	Rest	Hard exercise
Skeletal muscle	1000	20000
Kidneys	1100	600
Gut	1400	600
Brain	750	750

1. Calculate how many times muscle flow increases.
2. Explain the change for the kidneys and gut.
3. Explain why brain flow does not change.

Check-in: class 3

D3.3.11

1. State which organ receives the biggest increase in blood flow in exercise.
2. Explain how arterioles change blood flow.
3. Name two signals that cause vasodilation in active muscle.
4. Explain why blood flow to the brain stays constant.

How to use this

Answer without your notes, then check against slides 19–22.

Common mistakes to avoid

Not this	But this
Faeces are excretion.	Faeces are egestion; urea in urine is excretion.
Glucose is not filtered.	It is filtered, then fully reabsorbed.
ADH makes more urine.	ADH means less, more concentrated urine.
Blood vessels to muscles get wider by stretching.	Arteriole smooth muscle relaxes: vasodilation.

Check that you can...

1. **Distinguish** excretion and osmoregulation
2. **Explain** ultrafiltration in the glomerulus
3. **Explain** selective reabsorption in the proximal tubule
4. **Explain** the role of the loop of Henle
5. **Explain** how ADH controls water reabsorption by negative feedback
6. **Explain** how blood flow to organs changes with activity

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Explain how the glomerulus produces a filtrate.
2. Explain how ADH reduces water loss.
3. State two changes in blood flow during exercise.

Image credits

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FAMU athlete Robert Hayes practices running on the track.jpg – Florida Memory – Public domain
Human-Infrared.jpg – NASA/IPAC – Public domain

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